ADHITHYAN S

Penetration Tester | CPENT v2 & CEH v13 Certified

Bangalore, Karnataka • adhithyansatheesh2004@gmail.com • +91 9188671109 • 4-dhi.github.io • linkedin.com/in/adhithyan-s-pentester • github.com/4-dhi

PROFESSIONAL SUMMARY

Penetration Tester (CPENT v2, CEH v13) with hands-on, evidence-documented experience in Active Directory exploitation, binary exploitation, firmware reverse engineering, web application testing, and Linux privilege escalation. Builds and ships security tooling: VulnProbe (16-module web vulnerability scanner), WebRecon (15-module recon tool), and WebShield.AI (AI phishing-detection extension live on the Chrome and Edge stores). Seeking a penetration testing, red team, or web application security role.

TECHNICAL SKILLS

Penetration Testing Methodology: OWASP Top 10, SQL Injection, XSS, IDOR, SSRF, CSRF, RCE, Authentication Bypass, API Penetration Testing, JWT/OAuth2 Security, CVSS Risk Scoring, Evading Defense Mechanisms, Report Writing

Active Directory & Windows: LDAP enumeration, Kerberos AS-REP roasting, RODC assessment, Advanced Windows Attacks, credential/hash extraction (Impacket, NetExec), privilege escalation, credential cracking (John the Ripper, hashcat)

Binary Exploitation: Dynamic debugging and crash analysis (GDB), static disassembly/decompilation (Ghidra), exploit development and weaponization with pwntools, cyclic pattern offset analysis, exploit mitigation detection (checksec)

Web Application & CMS: Burp Suite, sqlmap, Gobuster, Joomscan, WPScan, Hydra

Firmware / IoT: Binwalk, SQLite3, file system and embedded credential extraction

Recon & Vulnerability Assessment: Subfinder, Amass, FFUF, Dirsearch, Nuclei, passive/active reconnaissance

Linux Privilege Escalation: SUID binary abuse, Linux capability abuse, sudo misconfiguration

Scripting & Automation: Python (security automation, attack automation, ML), Shell scripting, JavaScript

Cloud & Compliance: ISO/IEC 27001:2022 ISMS, IAM fundamentals, risk assessment

Platforms & Infrastructure: Kali Linux, Linux CLI, Pivoting and Double Pivoting (SSH SOCKS), Bypassing Filtered Networks, Manifest V3 extension development, REST API security, Nmap, Netcat

CERTIFICATIONS

- **Certified Penetration Testing Professional (CPENT) v2.** EC-Council. Issued Oct 2026, valid through Nov 2027.
- **Certified Ethical Hacker (CEH v13).** EC-Council
- **Advanced Diploma in Cyber Defense (ADCD).** Red Team Hacker Academy
- **ISO/IEC 27001:2022 Information Security Associate.** SkillFront
- **Cybersecurity Job Simulation.** Deloitte, via Forage
- **Cybersecurity Analyst Job Simulation.** TCS, via Forage. IAM and platform integration focus
- **CTF Achievement.** Technovally Hackfest 1.0
- **Advent of Cyber.** TryHackMe. Credential ID THM-LWPOOI9VLR

CPENT PRACTICAL EXAMINATION: TECHNICAL SUMMARY

- **Active Directory:** LDAP enumeration, AS-REP roasting (Impacket), hash cracking (John, hashcat), RODC credential exposure. 11/12 completed.
- **Binary Exploitation:** Crash analysis & exploit development with GDB, pwntools, Ghidra; built a working remote buffer-overflow exploit. 9/9 completed.
- **IoT / Firmware:** Reverse-engineered six firmware images (Binwalk, SQLite3), recovering embedded credentials and system details. 13/13 completed.
- **CTF-Style Linux Range:** SQL injection, file-upload RCE, CMS enumeration, privilege escalation (SUID, sudo, capabilities) across three hosts. 13/13 completed.
- **Web Application:** Access-control and logic-flaw exploitation. 2/6 completed.

KEY PROJECTS

VulnProbe: Web Application Vulnerability Scanner (Python, Flask) — 2026

- Built an authorization-first DAST scanner with 16 pluggable modules: reflected XSS, error-based and time-based blind SQLi, OS command injection, SSRF, path traversal, CORS, CSRF, open redirect, clickjacking, host header, sensitive file and header/cookie checks
- Engineered a scope and safety core that validates every request and redirect hop, blocks private/loopback targets via DNS resolution, rate-limits traffic, and supports authenticated scans with secret redaction
- Added confidence scoring (confirmed/probable/unverified) with OAST token correlation to cut false positives, JS/OpenAPI endpoint discovery, Nuclei and OWASP ZAP adapters, PDF/JSON reports, and a 170-test pytest suite

WebShield.AI: AI-Powered Phishing Detection & Security Extension — 2025 – Present

Founder & Sole Developer | Chrome Web Store & Microsoft Edge Add-ons Store

- Engineered and deployed a Manifest V3 extension for real-time phishing detection, live on Chrome Web Store and Microsoft Edge Add-ons
- Trained a Random Forest model on URL and domain reputation signals to classify malicious URLs in real time (Python backend, PHP API)
- Identified and engineered defenses against interstitial/popup-based attacks and tracker-based data exfiltration via query-parameter stripping

WebRecon: Python Web Reconnaissance Tool — Open Source

github.com/4-dhi/-4-dhi-WebRecon

- Automated the pentest recon phase into one 15-task tool: footprinting (WHOIS, DNS, SSL/TLS, robots.txt, sitemaps), application analysis (security headers, server fingerprinting), and infrastructure mapping (WAF detection, directory enumeration, proxy/SSRF checks)
- Built with Python 3, Requests, BeautifulSoup, Python-Whois, WAFW00F, and LXML; validated end-to-end against OWASP Juice Shop, flagging missing security headers, exposed configuration-style paths, and proxy details

EDUCATION

Bachelor of Computer Application (BCA), University of Calicut | 2022 – 2025

Higher Secondary (Science), Memunda HSS | 2020 – 2022